

BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet

Atomic Structure

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: particles, nuclide notation, isotopes, A_r and ions.

A Retrieval

1. State the relative charge and relative mass of a proton, a neutron and an electron.

2. Define atomic number (Z) and mass number (A).

3. What is the same, and what is different, in a pair of isotopes?

B Knowledge

4. A neutral atom of ^{39}K has atomic number 19. Calculate the numbers of protons, neutrons and electrons.

Working

5. Explain why atoms are electrically neutral even though they contain charged particles.

C Application and calculation

6. Complete the table for ^{24}Mg , $^{24}\text{Mg}^{2+}$, ^{19}F and $^{19}\text{F}^{-}$.

Species	Protons	Neutrons	Electrons

7. Gallium occurs as 60.0% ^{69}Ga and 40.0% ^{71}Ga . Calculate A_r . Show substitution.

Working

8. Explain why the A_r of chlorine on the periodic table is not a whole number.

D Exam-style practice

9. A student claims that ^{14}C and ^{14}N are isotopes of each other. Explain why this is incorrect. [2]

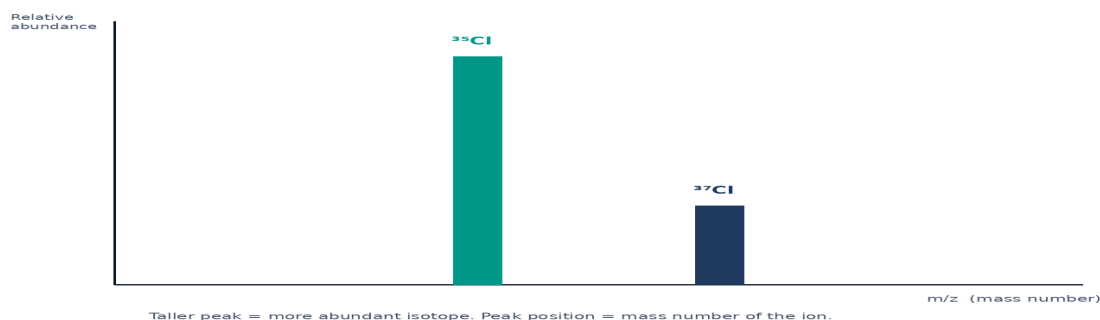
10. Calculate A_r for an element that is 75% isotope-63 and 25% isotope-65. [2]

Working

11. Explain how a 2^- ion forms from a neutral atom, referring to protons and electrons. [3]

E Challenge

Simple mass spectrum / abundance sketch



12. The sketch shows a chlorine-style abundance plot. A sample is 75.0% ^{35}Cl and 25.0% ^{37}Cl . Calculate A_r to 3 s.f. and explain why the value is not a whole number.

Working

13. On the blank nuclide card, complete A, Z, protons, neutrons and the electron number of Na^+ for sodium-23.

Nuclide notation · complete the labels

?

?

Na

Mass number A =

Atomic number Z =

Protons =

Neutrons =

Electrons in Na^+ =
